

Digital Material Passport

ID **DMP-METAL-005** - Version **1.0.0**
Issue Date **2025-05-17** - Certificate Type **EN 10204 3.1**

Manufacturer		Customer		Business Transaction				
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com		European Research Institute Science Boulevard 42 75015 Paris FR materials@eri.example.org		Order	PO-23456 · 1000 kg Pos. 1 · 2025-05-01			
				Delivery	DN-65432 · 1000 kg Pos. 1 · 2025-05-16 Internal Order ID: SO-54321			
Product								
Product Name	Titanium Alloy Ti-6Al-4V ELI			Number (UNS)	R56401			
Batch ID	T-65432-01			Shape	RoundBar - D 30 mm - Length 3000 mm			
Surface Condition	Machined			Product Norm	ASTM F136 (2022)			
Weight	1000 kg							
Production Date	2025-05-15							
Country of Origin	DE							
Chemical Analysis								
Heat Number T-65432 - Melting Process VAR - Casting Method VacuumCasting - Casting Date 2025-05-14 - Sample Location Product								
Symbol	Ti	Al	V	Fe	O	C	N	H
Unit	%	%	%	%	%	%	%	ppm
Min		5.50	3.50					
Max		6.50	4.50	0.25	0.13	0.080	0.050	120
Actual	89.32	6.02 ± 0.05	3.95 ± 0.03	0.18	0.11	0.026	0.012	35
Mechanical Properties								
Property	Actual			Minimum	Maximum	Method	Status	
Fatigue Test Room temperature, R = 0.1						ASTM E466	✓	
Cycles (N)	10000			50000	100000	500000	1000000	10000000
Value	650			635	610	590	570	550
Notch Sensitivity	0.85 - 0.92					Internal Method TS-5432	✓	
Physical Properties								
Property	Actual			Target/Min	Maximum	Method	Status	
Density	4.43 ± 0.01 g/cm³			4.43 g/cm³		ASTM B311	✓	
Supplementary Tests								
Test	Result / limits				Method	Status		
Microstructure Examination	Equiaxed alpha with intergranular beta				ASTM E407	✓		
Ultrasonic Inspection	Yes - No indications greater than reference standard				ASTM E2375	✓		
Surface Quality Assessment	Class 1 - Medical Grade				Visual Inspection per ASTM F136	✓		
Alpha Case Depth	5 µm - max 25 µm				Microhardness Traverse	✓		
Grain Size Distribution	8 - 10 ASTM No. - target 7 - 12 ASTM No.				ASTM E112	✓		
Hardness Profile (ASTM E384), values in HV								
Distance from surface (mm)	0.1	0.5	1	2	3	5	10	15
Value [HV]	345	350	352	350	349	348	351	347
Validation								
We hereby certify that the material described above has been manufactured and tested in accordance with ASTM F136 and meets all requirements for surgical implant applications.					Validated by Dr. Markus Weber , Head of Metallurgy, Research & Quality - 2025-05-17			